Example 1: Let $A=\begin{bmatrix}4&0&5\\-1&3&2\end{bmatrix}$, $B=\begin{bmatrix}1&1&1\\3&5&7\end{bmatrix}$, $C=\begin{bmatrix}2&-3\\0&1\end{bmatrix}$

Then, $A+B=\begin{bmatrix}5&1&6\\2&8&9\end{bmatrix}$. A is a 2×3 matrix and C is a 2×2 matrix. Since, they are not of the same size, $(A+C)$ is not defined. □

If r is a scalar, and A is a matrix, then the $\boxed{\text{scalar multiple}}$ rA is the matrix whose columns are r times the corresponding columns in A. As with vectors, $-A$ stands for $(-1)A$, and $A-B$ is the same as $A+(-1)B$.

Example 2: A and B are the matrices in Example 1, then

$$2B=2\begin{bmatrix}1&1&1\\3&5&7\end{bmatrix}=\begin{bmatrix}2&2&2\\6&10&14\end{bmatrix}$$

$$A-2B=\begin{bmatrix}4&0&5\\-1&3&2\end{bmatrix}-\begin{bmatrix}2&2&2\\6&10&14\end{bmatrix}=\begin{bmatrix}2&-2&3\\-7&-7&-12\end{bmatrix}$$

Theorem 1: Let A, B and C be matrices of the same size, and let r and s be scalars.

a. $A+B=B+A$
b. $(A+B)+C=A+(B+C)$
c. $A+0=A$
d. $r(A+B)=rA+rB$
e. $(r+s)A=rA+sA$
f. $r(sA)=(rs)A$

Proof: Let A, B and C be matrices of size $m\times n$.

$A=[a_1\ a_2 \cdots\cdots a_n]$, $B=[b_1\ b_2 \cdots\cdots b_n]$ $C=[c_1\ c_2 \cdots\cdots c_n]$

a) $A+B=[a_1+b_1\ \ a_2+b_2 \cdots\cdots a_n+b_n]=[b_1+a_1\ \ b_2+a_2 \cdots\cdots b_n+a_n]$ (Algebraic property (i))

$=B+A.$

b) $(A+B)+C=([a_1\ a_2 \cdots\cdots a_n]+[b_1\ b_2 \cdots\cdots b_n])+[c_1\ c_2 \cdots\cdots c_n]$

$=[(a_1+b_1)\ (a_2+b_2) \cdots\cdots (a_n+b_n)]+[c_1\ c_2 \cdots\cdots c_n]$

$=[(a_1+b_1)+c_1\ \ (a_2+b_2)+c_2 \cdots\cdots (a_n+b_n)+c_n]=[a_1+(b_1+c_1)\ \ a_2+(b_2+c_2) \cdots\cdots a_n+(b_n+c_n)]$ (Algebraic property (ii))